Problem Set 13

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: MML 8.2 (empirical risk minimization, regularization, cross-validation); ISL 5 (resampling and cross-validation). Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited exercise numbers are from ISL, 2nd edition (James et al., 2021).

1. ISL, Chapter 5, Conceptual exercises: problems 1, 2, 3.
2. Explain, in a short paragraph, the bias–variance trade-off and how it relates to overfitting. Give one concrete example of an under-fit model and one of an over-fit model for the same data.
3. **[code]** Fit polynomials of increasing degree to a small data set. Use k-fold cross-validation to select the degree, and plot training error versus validation error as a function of degree.
4. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).